
5.6 INITIAL CAL/BIT AND TOA DELAY SETTINGS

INITIAL SETUP:

Tools and Special Tools

920-00636-100, Site Calibration Kit

References

5 System Configuration WP
6.2 Calibration WP

Materials/Parts

None

Equipment Condition

5.5 Input Survey Data WP Completed
Equipment Powered On
Generator or Shore power applied

Personnel Required

Technician (2)

GENERAL INFORMATION

- This procedure is designed to set initial Cal/BIT and frequency settings. Before TOA delay settings can be calculated, survey data must be entered, and system must be passing BIT.
 - The following procedure will need to be repeated for the second rack if system is configured with dual racks.
 - Switch to the rack being configured on the Remote Control Unit (RCU).
 - This subsection deals with positioning the Cal/BIT pulse in the “framing” window. This will allow the system to take measurements from the pulse to ensure the health of the system. The windowing correlates to the physical distance between the Cal/BIT and the receive-antennas. After the initial power on of the system this is the first step that is required to be completed to set the monitor limits of the system. The following procedure will require that the system is installed to the siting requirements, and the system survey is completed, and all relevant information is entered to the system software.
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Framing the Cal/BIT TOA Window

1. Press “Archive...” button on MI.

NOTE

Do not change the default file name.

2. Press “Run” button in Archive Dialog.
3. Press “Close” when process is complete to return to MI main window.
4. In SYSTEM SETUP>SETTINGS> DATA RECORDING, ensure “Record BIT RAM files” is checked (See Figure 6-1).
5. Select OK.
6. Close system setup window.

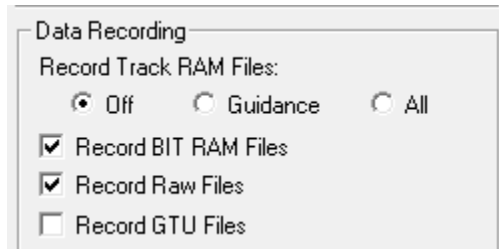


Figure 6-1. Record BIT RAM Files

7. Select “Start System” to run Built-In-Test.

NOTE

“Base 2 Run Folder,” has a shortcut on the desktop.

8. From the Run folder on base2, open the “asabram” folder (base2\run\raw\asabram) and open the most recent *.b.ram file.

NOTE

Adjustments will be made in the SYSTEM SETUP>MONITOR LIMITS, in either the (ESA, ASA, or ATA) (See Figure 6-2).



Figure 6-2. TOA Bias Setting

9. In the RAM tool ensure the Reference (ASA) Medium (ATA) channel are selected (See Figure 6-3).

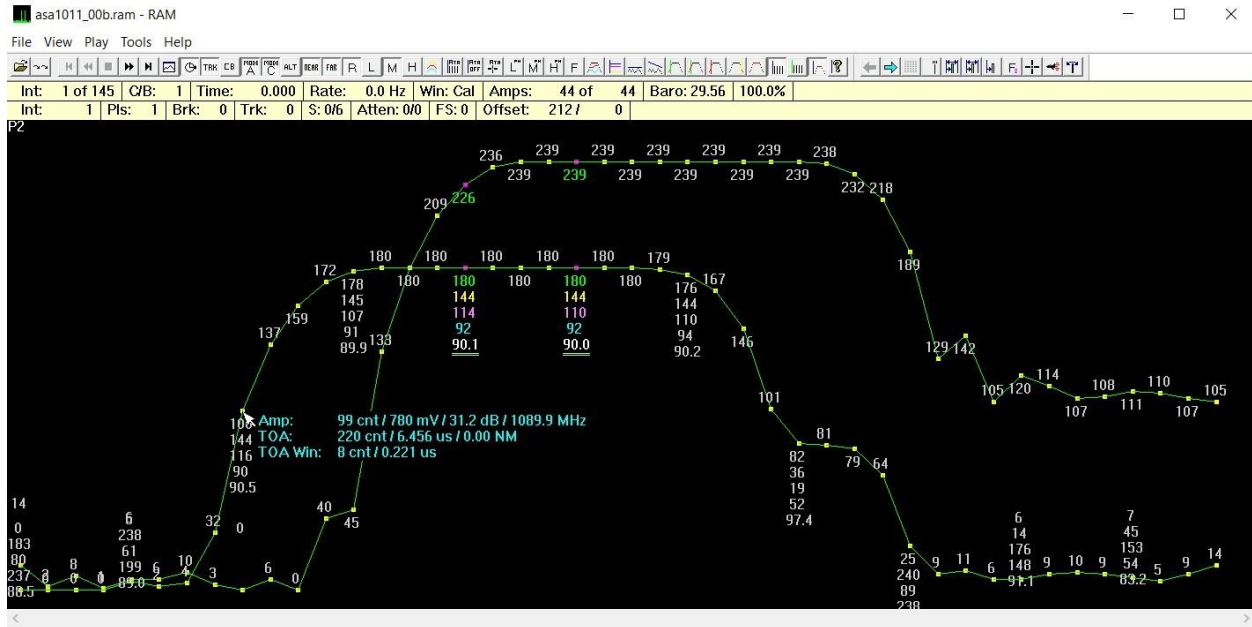


Figure 6-3. ASA and ATA Channels Selected

NOTE

- The ASA and ATA pulses won't be perfectly aligned, this procedure will be considered successful when both the ASA and ATA pulses are viewable in the window.
- ASA channels are Reference, Low and High.
- ESA channels are Reference, Low, Medium, and High.
- The ATA pulse is visible on the medium channel of the ASA b.ram file.
- The goal is to center the pulse so that entirety of the pulse is in the window.
- Use following steps to bring Cal/BIT level within measurement limits of TLS AOA sensors.

10. If pulse is not centered in the window (See Figure 6-3), perform the following:
 - a. Open MI.
 - b. Stop System Software.
 - c. Select Setup.
 - d. Select Monitor Limits.
 - e. Select ASA Limits
 - f. Under TOA, input information in Near Field.

- i. If the pulse is visible in the RAM window, hold “Shift” and click on approximately the midpoint of the leading edge of the ASA Ref pulse. This will display a dialogue box with the TOA near bias listed in μs (See Figure 6-3). Convert this number to ns ($6.456 \mu\text{s} = 6,456 \text{ ns}$) and enter in the Near field.
 - ii. If there is no pulse in the window, first decrease the TOA Bias by 300ns to see if the pulse is closer in time, if not then increase setting to check if the pulse is farther out in time.
- g. Once the TOA Bias has been adjusted, select OK.
- h. Select Monitor Limits.
- i. Select ATA Limits
- j. Under TOA, input information in Near Field.
- k. Ensure the ASA and ATA TOA Biases are adjusted symmetrically. For example, if you decrease the TOA Bias for the ASA by 300ns, you will want to decrease the ATA TOA bias by 300ns as well.
- l. Once the TOA Bias has been adjusted, select OK.
- m. Start the system software to produce another asa*b.ram file.
- n. Examine the Reference and Medium pulses to ensure TOA bias changes have taken effect. A centered pulse is shown in Figure 6-4.

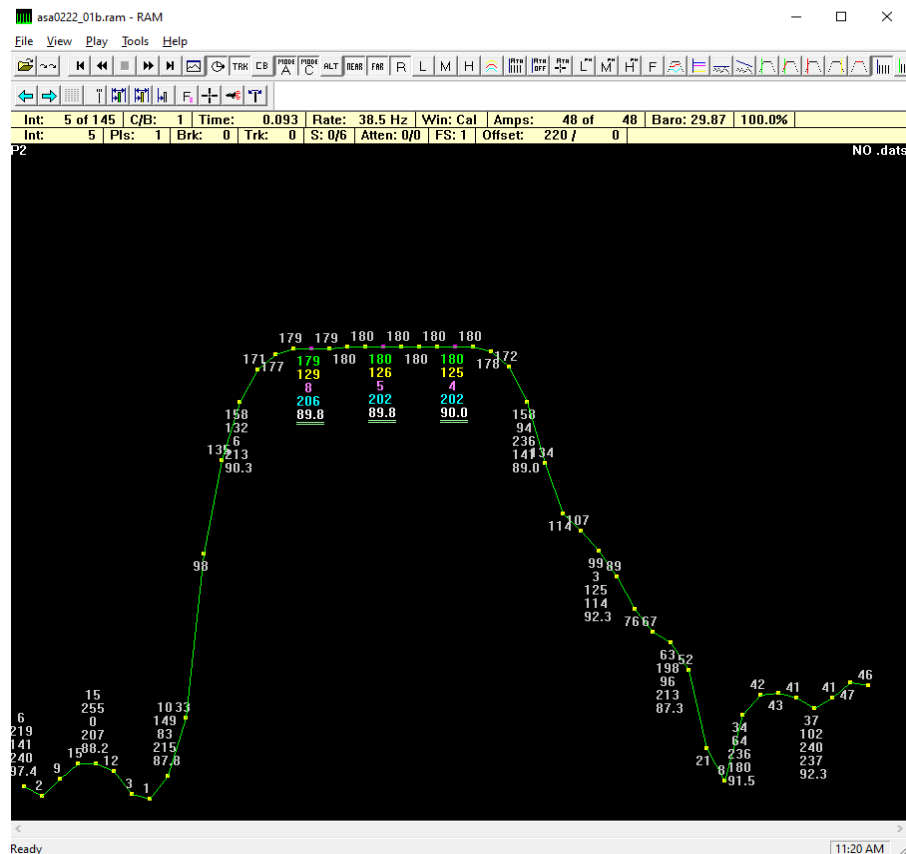


Figure 6-4. Cal/BIT Centered in Window

11. Repeat above steps for the ESA REF channel using the esa*b.ram file.

NOTE

After the process is complete the *.bit files will fill with data points as BIT is ran. This is critical for the system to then create the base line in a later procedure.

Cal/BIT Power Settings

NOTE

- Use following steps to bring Cal/BIT level within measurement limits of TLS sensors.
 - ASA channels are Reference, Low and High.
 - ESA channels are Reference, Low, Medium, and High.
 - ATA channel is Medium.
1. On MI, select Tool.
 2. Review BIT Monitor Data (MultiBIT).
 3. Select File
 4. Select ASA BIT
 - a. Select One Plot (Light Switch)
 - b. Select Plot Dots (3 Dots)
 - c. Select Reference Amplitude (R)
 5. Repeat previous steps 1 through 4c for ESA and ATA.

NOTE

- There should be approximately 75 mV of change for each 3 dB change in attenuation.
 - Channels for either sensor must have Cal/BIT amplitude measurements between 1000mV to 1950mV.
6. Ensure that power is removed from Cal/BIT transmitter before disconnecting RF and adjust attenuations as follows:
 - a. Ensure power to Cal/BIT is removed by stopping system software.
 - b. Open the Cal/BIT electronics enclosure at antenna base.

- c. Add or remove attenuators as needed to adjust.
 - d. Close and secure cover.
7. Start system software on MI.

NOTE

Entering and exiting test mode once will get 3 BIT cycles to use as a measure of the new Cal/BIT amplitude.

8. Enter and exit Test mode.
9. On MI, select Tool.
10. Select review BIT Monitor Data
11. Select File
12. Select ASA BIT
 - a. Select One Plot (Light Switch)
 - b. Select Plot Dots (3 Dots)
 - c. Select Reference Amplitude (R)
13. Verify the following:
 - a. There should be approximately 75 mV of change for each 3 dB change in attenuation.
 - b. Channels for either sensor must have Cal/BIT amplitude measurements between 1000mV to 1950mV.
14. Repeat steps 9 through 12c for ESA and ATA BITs to verify amplitudes are within range.
15. Determine if adjustments are required for ASA, ESA and ATA BITs.
 - a. If additional adjustments are required after checking each BIT, repeat steps 6 through 8 as necessary followed by the BIT examination steps 9 through 12c for each BIT to verify proper amplitude.
16. Stop system software.

TOA F1 Tuning

NOTE

- The following steps are completed to obtain the most stable Cal/BIT pulse.
- "T_i" refers to the current F1 Cal/Bit Position before beginning this process and comes from the factory with system default of 5 μ S.
- If procedure is being done as part of a replacement of an LRU, "T_i" refers to current nominal setting for F1 Cal/BIT Delay.